

must use a more general calculation (see Chapter 16 for the derivation). But even in the homoskedastic case, we are not finished. The formula

$$\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$$

still contains σ^2 , an unknown population quantity. We cannot plug in what we do not know. So we need to estimate σ^2 from the data. Fortunately, the hat matrix from Chapter 9 gives us everything we need.

Before we move on, though, let's pause and look at what $\text{Var}(\hat{\beta})$ actually means in practice. Figure 10.2 draws four samples from the exact same population. Same true slope, same true intercept, same σ . Yet every sample gives a wildly different OLS line. One sample (panel a) lands almost on top of the truth. Another (panel b) is twice as steep. Panel c finds a slope that is barely positive, and panel d gets the sign completely wrong. All four came from the *same* data generating process. That is sampling variation, and $\text{Var}(\hat{\beta}) = \sigma^2(X'X)^{-1}$ is the formula that tells us how much we should expect those lines to wiggle from sample to sample.

10.2 Estimating σ^2 and Calculating Standard Errors

With the variance formula in hand and the properties of H and $I_n - H$ from Chapter 9, we can derive the formula for estimating σ^2 and, from there, the standard errors.

The full chain from residuals to standard errors is built in seven steps. We lay them out one at a time so the logic is clear.

Step 1: Put $\hat{u}$ in terms of $I_n - H$ and $\mathbf{u}$

$$\begin{aligned} \hat{\mathbf{u}} &= (I_n - H)\mathbf{y} && \text{Definition of residuals} \\ &= (I_n - H)(X\beta + \mathbf{u}) && \text{Substitute } \mathbf{y} = X\beta + \mathbf{u} \\ &= \underbrace{(I_n - H)X\beta}_{=0} + (I_n - H)\mathbf{u} && \text{Distribute; from (9.3) the first term vanishes} \\ &= (I_n - H)\mathbf{u} && \text{Signal vanishes, only noise remains} \end{aligned}$$

Step 2: Sum of squared residuals. Now take the inner product $\hat{\mathbf{u}}'\hat{\mathbf{u}}$ and simplify:

$$\begin{aligned}
\hat{\mathbf{u}}'\hat{\mathbf{u}} &= \mathbf{u}'(I_n - H)'(I_n - H)\mathbf{u} && \text{Substitute } \hat{\mathbf{u}} = (I_n - H)\mathbf{u} \\
&= \mathbf{u}'(I_n - H)^2\mathbf{u} && \text{Symmetric (Section 2.8.2): } (I_n - H)' = I_n - H \\
&= \mathbf{u}'(I_n - H)\mathbf{u} && \text{Idempotent (Section 2.8.2): } (I_n - H)^2 = I_n - H
\end{aligned} \tag{10.3}$$

Step 3: Connecting $\mathbb{E}[\mathbf{u}\mathbf{u}' \mid X]$ to $\sigma^2 I_n$. The next step will produce the expression $\mathbb{E}[\mathbf{u}\mathbf{u}' \mid X]$, and we are going to replace it with $\sigma^2 I_n$. That replacement is not obvious, so let's prove it.

First, what is $\mathbf{u}\mathbf{u}'$? It is the **outer product** of the error vector with itself: an $n \times n$ matrix whose (i, j) entry is $u_i u_j$. Taking its expectation gives us a matrix whose (i, j) entry is $\mathbb{E}[u_i u_j \mid X]$. We need to connect this matrix to the variance-covariance matrix $\text{Var}(\mathbf{u} \mid X)$ that we already know from Assumption 5.

In Chapter 2, we defined the variance-covariance matrix of a random vector $\mathbf{z}$ as:

$$\text{Var}(\mathbf{z}) = \mathbb{E}[(\mathbf{z} - \mathbb{E}[\mathbf{z}])(\mathbf{z} - \mathbb{E}[\mathbf{z}])'].$$

This is the matrix version of the scalar formula $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$. Let's expand it for our error vector $\mathbf{u}$ (conditioning on X throughout). Write $\boldsymbol{\mu} = \mathbb{E}[\mathbf{u} \mid X]$ to keep things compact:

$$\begin{aligned}
\text{Var}(\mathbf{u} \mid X) &= \mathbb{E}[(\mathbf{u} - \boldsymbol{\mu})(\mathbf{u} - \boldsymbol{\mu})' \mid X] && \text{Definition from Chapter 2} \\
&= \mathbb{E}[\mathbf{u}\mathbf{u}' - \mathbf{u}\boldsymbol{\mu}' - \boldsymbol{\mu}\mathbf{u}' + \boldsymbol{\mu}\boldsymbol{\mu}' \mid X] && \text{FOIL the product} \\
&= \mathbb{E}[\mathbf{u}\mathbf{u}' \mid X] - \boldsymbol{\mu}\boldsymbol{\mu}' - \boldsymbol{\mu}\boldsymbol{\mu}' + \boldsymbol{\mu}\boldsymbol{\mu}' && \text{Take } \mathbb{E} \text{ of each term; } \boldsymbol{\mu} \text{ is constant given } X \\
&= \mathbb{E}[\mathbf{u}\mathbf{u}' \mid X] - \boldsymbol{\mu}\boldsymbol{\mu}' && \text{Last two terms cancel}
\end{aligned}$$

Rearranging:

$$\mathbb{E}[\mathbf{u}\mathbf{u}' \mid X] = \text{Var}(\mathbf{u} \mid X) + \boldsymbol{\mu}\boldsymbol{\mu}'.$$

The zero conditional mean assumption (Assumption 3) says $\boldsymbol{\mu} = \mathbb{E}[\mathbf{u} \mid X] = \mathbf{0}$. So $\boldsymbol{\mu}\boldsymbol{\mu}' = \mathbf{0} \cdot \mathbf{0}' = \mathbf{0}$, and we are left with:

$$\mathbb{E}[\mathbf{u}\mathbf{u}' \mid X] = \text{Var}(\mathbf{u} \mid X).$$

When the mean is zero, the average of the square *is* the variance. Finally, homoskedasticity (Assumption 5) tells us $\text{Var}(\mathbf{u} \mid X) = \sigma^2 I_n$, so:

$$\mathbb{E}[\mathbf{u}\mathbf{u}' | X] = \sigma^2 I_n.$$

Step 4: Expected value. Taking the conditional expectation of both sides of Equation (10.3):²

$$\begin{aligned}
 \mathbb{E}[\hat{\mathbf{u}}'\hat{\mathbf{u}} | X] &= \mathbb{E}[\mathbf{u}'(I_n - H)\mathbf{u} | X] && \text{Equation (10.3)} \\
 &= \mathbb{E}[\text{tr}(\mathbf{u}'(I_n - H)\mathbf{u}) | X] && \text{A scalar equals its trace}^3 \\
 &= \mathbb{E}[\text{tr}((I_n - H)\mathbf{u}\mathbf{u}') | X] && \text{Cyclic property of trace} \\
 &= \text{tr}(\mathbb{E}[(I_n - H)\mathbf{u}\mathbf{u}' | X]) && \mathbb{E}[\text{tr}(A)] = \text{tr}(\mathbb{E}[A]) \\
 &= \text{tr}((I_n - H) \mathbb{E}[\mathbf{u}\mathbf{u}' | X]) && (I_n - H) \text{ is constant given } X \\
 &= \text{tr}((I_n - H)\sigma^2 I_n) && \text{From Step 3, } \text{Var}(\mathbf{u} | X) = \sigma^2 I_n \\
 &= \sigma^2 \text{tr}(I_n - H) && \text{Pull out the scalar } \sigma^2 \\
 &= \sigma^2(n - \text{tr}(H)) && \text{tr}(I_n) = n \\
 &= (n - k)\sigma^2 && \text{tr}(H) = k \text{ (Section 9.2)}
 \end{aligned}$$

Step 5: The variance estimator. Rearranging gives us an unbiased estimator of σ^2 :⁴

$$\hat{\sigma}^2 = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n - k} \quad \mathbb{E}[\hat{\mathbf{u}}'\hat{\mathbf{u}} | X] = (n - k)\sigma^2; \text{ divide both sides by } (n - k) \quad (10.4)$$

The denominator $n - k$ is the **degrees of freedom** of the regression (see Section 3.6 for a more complete explanation). We start with n observations, but estimating k coefficients forces k linear restrictions on the data. After fitting the model, only $n - k$ independent pieces of variation remain in the residuals. Those remaining pieces are what we use to estimate σ^2 . Notice that we did not choose to divide by $n - k$ simply out of convention. The $n - k$ came of the derivation: Step 4 showed that $\mathbb{E}[\hat{\mathbf{u}}'\hat{\mathbf{u}} | X] = (n - k)\sigma^2$, so dividing by $n - k$ is the only choice that makes the estimator unbiased. We started with n observations and used up k of them estimating the coefficients, leaving $n - k$ independent pieces of information in the residuals.

² The hat matrix H is built entirely from X , so once we know X we know H . Conditioning on X lets us treat H as a known constant, and the only randomness left comes from the errors $\mathbf{u}$.

³ Any scalar equals its trace. Using the cyclic property of the trace (Section 2.5.2), $\text{tr}(ABC) = \text{tr}(BCA)$, we obtain

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \text{tr}(\mathbf{x}'\mathbf{A}\mathbf{x}) = \text{tr}(\mathbf{A}\mathbf{x}\mathbf{x}').$$

This identity is frequently used when working with quadratic forms.

⁴ The quantity $\hat{\sigma}^2$ is often called the **mean squared error** (MSE) of the regression.

The geometry behind $n - k$. The algebra above showed us *that* the degrees of freedom are $n - k$. The geometry of projection tells us *why*.

Recall from Chapter 9 that the residual vector $\hat{\mathbf{u}} = (I_n - H)\mathbf{y}$ is the orthogonal projection of $\mathbf{y}$ onto the space perpendicular to $\text{Col}(X)$. The column space of X is k -dimensional (one dimension for each regressor, including the intercept). Since $\mathbf{y}$ lives in $\mathbb{R}^n$ and $\text{Col}(X)$ occupies k of those n dimensions, the orthogonal complement has dimension $n - k$. That is the space the residuals are confined to. They cannot point in any direction they want; they are trapped in an $(n - k)$ -dimensional subspace.

In our earlier 3D example ($n = 3$, $k = 2$), the column space was a plane, and the residuals could only point in the one direction perpendicular to that plane: $3 - 2 = 1$ degree of freedom. With real data, say $n = 500$ and $k = 5$, the residuals live in a 495-dimensional space. More room, but still fewer dimensions than the original data.

This is what “degrees of freedom” *means* geometrically: it is the number of independent directions the residuals are free to move in. The algebra in Step 4 gave us $\text{tr}(I_n - H) = n - k$, which is the trace of the projection matrix onto this complement. The geometry tells us that trace counts exactly the number of dimensions of the subspace. Trace equals dimension equals degrees of freedom. These are three names for the same number, arrived at from three different directions. When we add the normality assumption in Chapter 12, this geometric fact will give us the *entire distribution* of the residual sum of squares, not just its expected value.

Step 6: Estimated variance of $\hat{\beta}$. Replace the unknown σ^2 with our estimate:

$$\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2 (X'X)^{-1} = \frac{\hat{\mathbf{u}}'\hat{\mathbf{u}}}{n - k} (X'X)^{-1} \quad \text{Replace unknown } \sigma^2 \text{ with } \hat{\sigma}^2 \text{ in (10.2)}$$

Step 7: Standard errors. Finally, take the square root of each diagonal element:

$$\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 \cdot [(X'X)^{-1}]_{jj}} \quad \text{Square root of } j\text{th diagonal element} \quad (10.5)$$

Notice that nothing in $I_n - H$ requires the vector being projected to be $\mathbf{y}$. If we apply $(I_n - H)$ to *any* $n \times 1$ vector $\mathbf{v}$, the result $(I_n - H)\mathbf{v}$ is the part of $\mathbf{v}$ orthogonal to $\text{Col}(X)$: the residuals from regressing $\mathbf{v}$ on X . We can build a residual maker for any set of variables. If Z is some subset of our control variables, then $I_n - Z(Z'Z)^{-1}Z'$ removes the influence of Z from whatever we apply it to. Want to “clean” both X and Y of the effect of Z ? Just apply the Z residual maker to each. This idea is the engine behind the **Frisch Waugh Lovell theorem** (Chapter 13), which shows that the coefficient on X in a multiple regression equals the slope from a simple regression after removing the influence of Z from both sides. The residual maker turns “controlling for other variables” into a concrete, mechanical operation.

For simple linear regression, the $(1, 1)$ element of $(X'X)^{-1}$ is $1/\sum(X_i - \bar{X})^2$, so:

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^n (X_i - \bar{X})^2}} \quad \text{SLR special case} \quad (10.6)$$

We would clearly like our standard errors to be as small as possible, but what could affect this? Look at formula (10.5) and think about what each piece does:

- **Small residuals** ($\hat{\mathbf{u}}'\hat{\mathbf{u}}$ is small): The model fits well. Little noise means less uncertainty about the slope.
- **Lots of variation in X** ($[(X'X)^{-1}]_{jj}$ is small): The explanatory variables spread out over a wide range, giving more information about the slope.
- **More degrees of freedom** ($n-k$ is large): More observations relative to parameters means more information for estimating σ^2 .

In short: **OLS loves spread in X** because it gives more information about the slope, and it **dislikes noisy residuals** because they obscure the signal.

The standard errors in practice

We just derived $\widehat{\text{Var}}(\hat{\beta}) = \hat{\sigma}^2(X'X)^{-1}$ and showed that the standard error of the j th coefficient is $\text{SE}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2[(X'X)^{-1}]_{jj}}$. Let's see what that formula actually produces. We will take the baseball data, compute the OLS residuals, estimate $\hat{\sigma}^2 = \hat{\mathbf{u}}'\hat{\mathbf{u}}/(n-k)$, form the full variance covariance matrix, and extract the standard errors from its diagonal. Every step follows directly from the derivation above.

	$\hat{\beta}$	$\widehat{\text{Var}}(\hat{\beta})$	SE
Constant	-4.9937	0.5193	0.7206
On Base Pct	19.1979	11.6295	3.4102
Slugging Pct	8.4121	2.5098	1.5842

$$s^2 = 0.0173$$

$$\widehat{\text{Var}}(\hat{\beta}) = s^2(X'X)^{-1} = \begin{pmatrix} 0.5193 & -2.0058 & 0.2799 \\ -2.0058 & 11.6295 & -4.1050 \\ 0.2799 & -4.1050 & 2.5098 \end{pmatrix}$$

Look at the magnitudes. The coefficient on On Base Percentage is 19.20 with a standard error of 3.41, so the ratio is about $19.20/3.41 \approx 5.6$. The coefficient on Slugging Percentage is 8.41 with a standard error of 1.58, giving a ratio of about $8.41/1.58 \approx 5.3$. Both coefficients are several times their standard errors, which is a sign that we have measured something real. We are not ready to conclude anything yet (that comes in Chapter 12), but the intuition is already here: a coefficient that is large relative to its

standard error is one we can be confident about, while a coefficient close to its standard error might just be noise.

Let’s return to Fisher’s 1925 data. We computed $\hat{\beta}$ earlier. Now we compute the standard errors and the t -statistic and compare against the exact numbers Fisher published in *Statistical Methods for Research Workers*, Table 29.

	$\hat{\beta}$	$\widehat{\text{Var}}(\hat{\beta})$	SE
Constant	0.3317	4.3100	2.0761
Trend	0.2668	0.0137	0.1169

$$s^2 = 30.7354$$

$$\widehat{\text{Var}}(\hat{\beta}) = s^2 (X'X)^{-1} = \begin{pmatrix} 4.3100 & -0.2120 \\ -0.2120 & 0.0137 \end{pmatrix}$$

10.3 The Gauss–Markov Theorem: Why OLS Is “Best”

We have shown that OLS is unbiased: on average, we should get the correct estimate of β . But unbiasedness alone is not enough to recommend an estimator. There are *many* unbiased estimators. For instance, you could pick any two observations from your sample and draw a line through them. That estimator would be unbiased too. But it would be wildly imprecise, because it throws away most of the data.

So here is the natural next question: among all the unbiased estimators we could come up with, is OLS the most precise one? Does it have the smallest variance?

The answer is yes, as long as we restrict attention to estimators that are **linear** and **unbiased**. This result is known as the **Gauss–Markov theorem**, and it is one of the most important results in econometrics. It is the reason OLS earns the “B” in **BLUE** (**B**est **L**inear **U**nbiased **E**stimator). Let’s prove it carefully, showing every step.

What we already know about OLS

Before we start the proof, let’s collect the facts we have established so far. The OLS estimator $\hat{\beta} = (X'X)^{-1}X'y$ has the following properties:

1. It is a **linear** function of the observable random vector y .
2. It is a **random vector** with a sampling distribution.
3. It is **unbiased**: $\mathbb{E}[\hat{\beta} | X] = \beta$ (Section 8.2.2).